MAPLE: Modality-Adaptive Prior-Guided Learning With Language-Conditioned Spatial Adaptation for Multimodal Remote Sensing Segmentation

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***Abstract*—Multimodal remote sensing segmentation benefits from complementary information across heterogeneous data sources, yet effectively leveraging such information while preserving the generalization ability of pretrained models remains challenging. In this article, we propose modality-adaptive prior-guided learning (MAPLE), a unified framework that adapts a frozen foundation backbone for multimodal learning via lightweight modality-aware mechanisms and language-guided priors. By introducing adaptive feature modulation and semantic guidance, the proposed approach effectively mitigates modality discrepancies and improves spatial prediction consistency. In addition, an adaptive reweighting strategy is employed to address class imbalance and enhance the learning of underrepresented categories. Extensive experiments on multiple benchmarks demonstrate that MAPLE achieves state-of-the-art accuracy with the smallest trainable footprint among the compared methods, while a comprehensive efficiency analysis of total parameters, floating-point operations (FLOPs), inference latency, and memory consumption further confirms a favorable accuracy–efficiency trade-off. The source code is available at https://github.com/shiroSamaNanoda/MAPLE.git**

***Index Terms*—Foundation model, multimodal fusion, parameter-efficient adaptation, remote sensing, semantic segmentation.**

I. Introduction

MULTIMODAL remote sensing semantic segmentation aims to achieve fine-grained pixel-level understanding by jointly exploiting heterogeneous data sources, such as optical imagery and digital surface models (DSMs). By integrating complementary spectral and geometric cues, multimodal approaches have demonstrated clear advantages over single-modality methods in complex urban scenes. However, despite recent progress in CNN–Transformer hybrid architectures [1], [2], existing methods still rely heavily on task-specific training with limited data scale, which restricts their generalization ability across diverse geographic distributions and sensor conditions. This limitation has motivated the adoption of large-scale pretrained foundation models to provide transferable visual representations. Nevertheless, adapting such models to downstream multimodal segmentation introduces an additional efficiency challenge: achieving strong task performance with minimal task-specific parameterization while maintaining practical inference costs. To characterize this trade-off, Fig. 1 shows the comparison of segmentation accuracy against adaptation cost, measured by the number of trainable parameters, while the bubble size indicates the inference-time graphics processing unit (GPU) memory footprint. Existing methods often exhibit a pronounced tradeoff between segmentation accuracy and adaptation efficiency. We use trainable parameters on the horizontal axis because modality-adaptive prior-guided learning (MAPLE) is specifically designed to reduce task-adaptation cost; complementary inference-side metrics, including total parameters, floating-point operations (FLOPs), latency, and GPU memory, are further analyzed in Section IV-G.

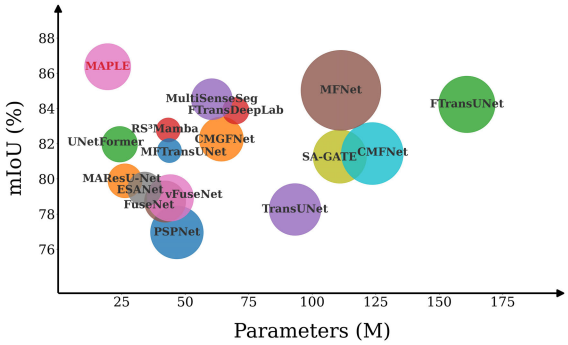


Fig. 1. Comparison of parameter efficiency, memory consumption, and segmentation performance on the ISPRS Vaihingen benchmark. The horizontal axis denotes the number of trainable parameters (in millions), the vertical axis shows mIoU (%), and the bubble size represents peak GPU memory usage during inference.

Recent advances in foundation models offer a promising direction. Building on the architectural evolution from SAM [3] through SAM2 [4], the recent SAM3 [5] integrates a perception encoder (PE) [6] backbone that provides improved representations for dense prediction tasks. Parameter-efficient adaptation strategies, such as Adapters [7] and LoRA [8], enable effective domain transfer with limited computational

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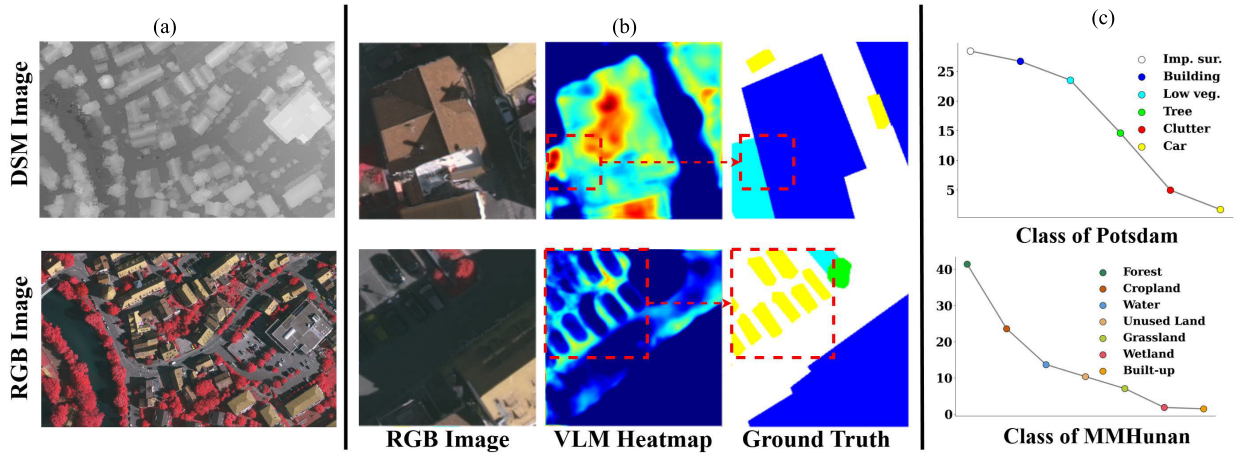


Fig. 2. Key challenges addressed by MAPLE. (a) Modality discrepancy: RGB imagery captures spectral appearance and texture, whereas DSM encodes complementary geometric and elevation information. (b) Limited spatial precision of vision-language models: patch-level semantic responses are spatially coarse and fail to align with fine-grained object boundaries, as illustrated by two representative cases for the *building* and *car* categories. The red dashed boxes highlight mismatches between the similarity heatmaps and the corresponding ground-truth labels. (c) Long-tailed class distribution: pixel-wise class proportions of the MMHunan and Potsdam datasets, where the horizontal axis denotes semantic categories (see legend), and the vertical axis represents the percentage of pixels.

overhead and have been successfully applied to SAM-based architectures in remote sensing [9], [10], [11]. Meanwhile, RemoteCLIP [12], a domain-adapted variant of CLIP [13] pretrained on remote sensing image-text pairs, offers semantically aligned features tailored to Earth observation data. In principle, combining such visual foundation models with vision-language semantic priors could significantly enhance multimodal segmentation. However, directly integrating these models into multimodal remote sensing segmentation remains non-trivial, due to three critical challenges.

First, foundation models such as SAM3 are pretrained on single-modality natural images, whereas multimodal remote sensing data exhibit substantial heterogeneity between spectral (RGB) and geometric (DSM) modalities [Fig. 2(a)]. While recent work such as MFNet [11] has begun to explore modality-specific fine-tuning, existing adaptation strategies remain largely content-agnostic: they apply the same transformation regardless of the local semantic context, and do not leverage external semantic priors to guide the adaptation process. Second, although vision-language priors (VLPs) provide rich high-level semantic knowledge, they are inherently optimized for global image-text alignment rather than dense prediction [Fig. 2(b)]. As a result, their patch-level responses are spatially coarse and often misaligned with fine-grained object boundaries. This creates a structural tension: while such priors are semantically informative, they are not directly suitable for precise spatial modeling. Third, remote sensing datasets typically exhibit severe long-tailed distributions at the pixel level [Fig. 2(c)]. Unlike instance-level imbalance in natural images, this imbalance manifests across dense spatial regions and varies within each mini-batch, making static global reweighting strategies insufficient.

To address these challenges, we propose a unified perspective: instead of treating foundation model priors as direct predictors, we leverage them as conditional signals to guide feature adaptation. Under this paradigm, semantically meaningful but spatially coarse priors can still play a critical role–not by determining precise outputs, but by modulating feature transformations at a region level, while fine-grained spatial localization is recovered downstream through dense supervision and decoder-level processing. This perspective separates MAPLE from three established directions. First, parameter-efficient adaptation methods [7], [8], [11], [14] learn task-specific transformations whose parameters remain fixed after training, without explicitly conditioning the adaptation on spatially resolved semantic priors. Second, CLIP-guided dense prediction [15], [16], [17] directly incorporates vision–language representations into feature construction or prediction, making performance dependent on their spatial localization quality, which is limited for encoders trained primarily with image-level objectives. Third, conventional multimodal fusion networks [1], [2], [18] commonly rely on modality-specific feature-processing branches and dedicated fusion modules, increasing task-specific capacity as additional modalities are incorporated. In contrast, MAPLE uses vision–language priors as input-dependent conditioning signals, leaves fine-grained localization to the segmentation backbone and decoder, and specializes both modalities on a single shared encoder through lightweight adapters and gating. MAPLE realizes this perspective through three components. In particular, we augment a frozen SAM3 backbone with language-conditioned modality-specific adapters (LMAs), whose transformations are dynamically modulated by semantic cues from VLPs to reduce modality discrepancies in a content-aware manner.

The main contributions of this work are summarized as follows.

1) We propose LMA, which uses vision–language priors as dense conditioning signals rather than direct predictors. The affine parameters of a bottleneck adapter are dynamically generated from RemoteCLIP similarity maps and factorized into semantic, spatial, and modality components, enabling position- and content-dependent adaptation with only ~28 K additional parameters.
2) We introduce a dual-branch VLP module built upon RemoteCLIP, which decouples semantic recognition and spatial localization into a similarity-based branch and a

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dense classification head (DCH), respectively, and integrates them through learnable per-class adaptive fusion.
3) We develop a spatial adaptive frequency reweighting (SAFR) loss that dynamically estimates classwise spatial frequency statistics with temporal smoothing via exponential moving average (EMA), providing more balanced patch-level supervision under the long-tailed class distributions prevalent in remote sensing datasets.
4) Extensive experiments on the Vaihingen, Potsdam, and MMHunan benchmarks demonstrate that the proposed MAPLE achieves state-of-the-art segmentation performance with 19.51 M trainable parameters on a single consumer-grade GPU, and we report a full efficiency profile including total parameters, FLOPs, latency, and memory for a balanced comparison.

II. Related Work

A. Multimodal Semantic Segmentation

The primary objective of multimodal semantic segmentation resides in the synergistic exploitation of heterogeneous sensor data to facilitate comprehensive pixel-level scene understanding. Initial research paradigms predominantly centered on feature-level integration within traditional encoder–decoder topologies. In particular, the introduction of self-attention-guided fusion in SA-Gate [18] enabled the adaptive weighting of modality-specific features, while ACNet [19] addressed multiscale cross-modal correlations through specialized attention complementary modules.

The subsequent transition toward Transformer-based architectures has significantly catalyzed the modeling of sophisticated cross-modal interactions. TokenFusion [20] and CMX [21] further explore cross-modal token exchange and feature rectification, while CMNeXt [22] generalizes to arbitrary sensor configurations. Other representative approaches, including FTransUNet [2] and CMFNet [1], further investigate Transformer-based fusion and cross-modal feature integration for remote sensing segmentation. Despite their effectiveness, these methods generally rely on trainable modality-specific encoders or fusion modules throughout the network, which can increase optimization and computational costs when extending a pretrained segmentation backbone to multimodal inputs.

B. Parameter-Efficient Foundation Model Adaptation

The adaptation of plain Vision Transformers for dense prediction tasks is primarily facilitated by the hybrid attention mechanisms within ViTDet [23], which harmonize windowed local attention with periodic global attention to maintain global context during high-resolution processing. This architectural paradigm is also adopted by the PE [6] used in SAM3 [5], providing a powerful pretrained representation for dense visual tasks. The efficacy of lightweight adaptation across natural and medical imaging domains has been extensively validated by SAM-Adapter [14] and Medical SAM Adapter [24]. Within the remote sensing context, RSPrompter [25] interrogated prompt-based adaptation for instance segmentation, while MFNet [11] established an early precedent for multimodal SAM extension through shared encoders and modality-specific fine-tuning on DSM data. These studies demonstrate that compact trainable components can effectively specialize a frozen or partially frozen foundation model. However, the learned adaptation parameters are generally fixed after training or determined primarily by the input modality, and therefore do not explicitly adapt to spatially varying semantic content. This limits the ability of conventional adapters to establish a direct connection between high-level semantic priors and low-level feature transformation.

C. VLPs in Dense Prediction

Adapting vision–language models for dense prediction presents a persistent structural bottleneck. While pixel-level representation extraction was advanced via contextual prompting in DenseCLIP [15], open-vocabulary segmentation necessitated the mask-based strategies explored in MaskCLIP [16] and CLIPSeg [17]. These studies demonstrate the potential of vision–language representations for providing semantic guidance beyond conventional visual-only supervision. However, as elucidated in SuperCLIP [26], the inherent contrastive pretraining objective of CLIP prioritizes global image–text alignment, frequently at the expense of fine-grained spatial correspondence, thereby inducing diffuse activations in patch-level applications. The potential for frequency-based reweighting to augment discriminative capability in vision–language learning was demonstrated in SuperCLIP, complementing alternative strategies predicated on region-level supervision [27] or extended long-text representations [28]. Collectively, these studies suggest that VLPs can provide useful semantic cues for dense prediction, but their role has predominantly centered on feature extraction or prediction itself, rather than on dynamically modulating the representation of a separate segmentation backbone.

D. Discussion

The above studies reveal two complementary directions for multimodal remote sensing segmentation. Multimodal fusion methods effectively exploit heterogeneous observations, but commonly depend on trainable modality-specific feature processing. Foundation-model adaptation provides a more parameter-efficient alternative, yet conventional adapters do not explicitly incorporate spatially resolved semantic information. Meanwhile, vision–language models offer transferable semantic priors, but are usually used as feature or prediction components rather than as a conditioning source for backbone adaptation.

MAPLE is motivated by this gap. Instead of directly using the vision–language model as a dense predictor, MAPLE exploits its output as a *dense semantic conditioning signal* for a shared frozen segmentation backbone. The resulting adaptation is spatially varying and modality-aware, allowing feature transformation to depend jointly on semantic context, spatial location, and input modality. This design connects the semantic knowledge of vision–language pretraining with parameter-efficient multimodal adaptation while avoiding the need for separate fully trainable modality-specific encoders.

III. Method

A. Overview

Given a paired RGB image I_{rgb} and a DSM I_{dsm}, MAPLE performs dense semantic segmentation using a frozen SAM3

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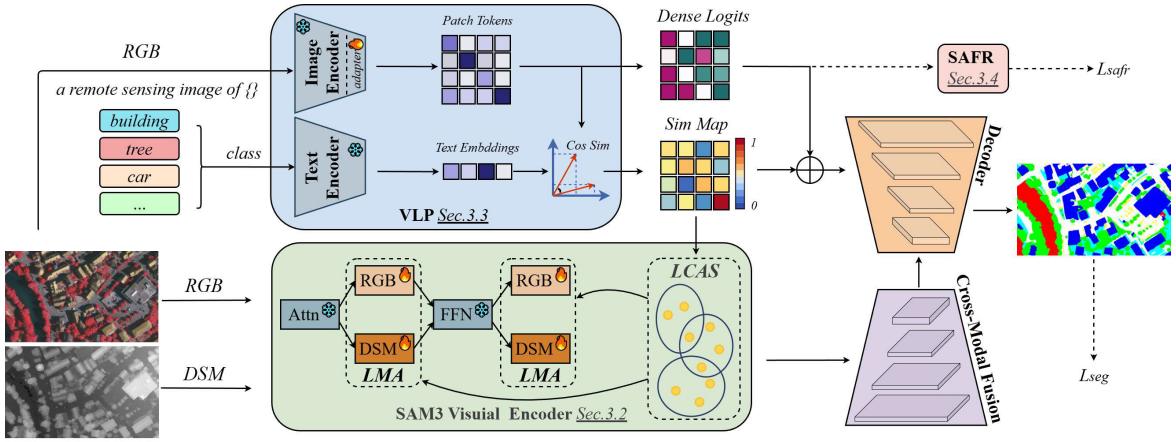


Fig. 3. Overall architecture of MAPLE. The SAM3 visual encoder (Section III-B) employs a shared frozen backbone for RGB and DSM inputs, augmented with LMA. These adapters leverage language-derived priors through LCAS, enabling content-aware and modality-differentiated feature adaptation. The VLP module (Section III-C) encodes visual patches and text prompts to produce classwise similarity maps and dense logits, which serve as the priors guiding the adaptation process. The resulting features are integrated via scalewise cross-modal fusion and decoded by a hierarchical global–local attention decoder. The final predictions are further refined by the VLP priors. During training, the SAFR loss (Section III-D) improves class balance by adaptively reweighting the dense logits, forming an end-to-end framework guided by language information.

backbone augmented with LMA (Section III-B), as illustrated in Fig. 3. The core idea is to leverage language-derived priors to guide multimodal feature adaptation in a content-aware manner.

To this end, we introduce a VLP (Section III-C) module based on RemoteCLIP, which produces classwise similarity maps and dense logits. These priors are used to condition LMA through the proposed language-conditioned adapter synthesis (LCAS), enabling position-aware and modality-differentiated feature modulation.

The adapted RGB and DSM features are then integrated through a standard cross-modal fusion scheme and decoded by a hierarchical architecture with global–local attention [29]. Finally, the model is optimized using a composite loss combining cross-entropy and the proposed SAFR (Section III-D) loss for improved class balance.

B. Parameter-Efficient Adaptation With LMAs

The MAPLE framework adopts SAM3 [5] as the visual backbone, leveraging its strong pretrained representations for dense prediction while keeping all backbone parameters θ_{frozen} fixed during training. Since the SAM3 backbone expects three-channel inputs while DSM is a single-channel modality, the DSM tensor is replicated along the channel dimension to form a pseudo-RGB input before being fed into the visual encoder, following the protocol adopted by MFNet [11] for the same purpose.

A fundamental challenge in multimodal remote sensing lies in the discrepancy between spectral and geometric modalities: RGB imagery captures radiometric appearance and fine-grained textures, whereas DSM encodes height-aware structural information. Uniform adaptation across modalities fails to account for such heterogeneity, leading to suboptimal representations.

To address this issue, we introduce LMA, lightweight bottleneck modules inserted after the self-attention and feed-forward layers of every transformer block of the frozen encoder. LCAS conditioning is activated at a selected subset of these blocks, as specified below. For modality m and transformer block l, the adapter output at spatial position (i, j) is

$$x'^{(l,m)}_{i,j} = x^{(l,m)}_{i,j} + \alpha W^{(l,m)}_{\text{up}} \left[\tilde{s}^{(l,m)}(i,j) \odot \text{GELU}\left(W^{(l,m)}_{\text{down}} x^{(l,m)}_{i,j} \right) + \tilde{h}^{(l,m)}(i,j) \right] \tag{1}$$

where $W^{(l,m)}_{\text{down}} \in \mathbb{R}^{r\times d}$ and $W^{(l,m)}_{\text{up}} \in \mathbb{R}^{d\times r}$ form a bottleneck projection from the backbone dimension d to the adapter dimension r and back to d. The learnable residual scaling factor α is initialized to 0.1. At LCAS-conditioned blocks, the affine modulation parameters $\tilde{s}^{(l,m)}(i,j), \tilde{h}^{(l,m)}(i,j) \in \mathbb{R}^r$ are generated jointly from semantic, spatial, and modality factors; at the remaining blocks, they reduce to $\mathbf{1}$ and $\mathbf{0}$, respectively.

1) Semantic Conditioning: For each LCAS-conditioned block $l \in \mathcal{L}_{\text{LCAS}}$ and semantic class c, we maintain a learnable raw scale parameter $a^{(l)}_c \in \mathbb{R}^r$ and shift parameter $h^{(l)}_c \in \mathbb{R}^r$. The effective semantic scale is defined as

$$s^{(l)}_c = 2\sigma\left(a^{(l)}_c\right). \tag{2}$$

The raw scales $a^{(l)}_c$ and shifts $h^{(l)}_c$ are initialized to zero, yielding $s^{(l)}_c = \mathbf{1}$ and $h^{(l)}_c = \mathbf{0}$ at initialization. Together with the small residual coefficient α and the near-zero initialization of W_{up}, this gives a near-identity initialization that preserves the pretrained backbone behavior at the beginning of optimization.

2) Spatial and Modality Conditioning: The normalized similarity map $\mathcal{S}_{\text{sim}}$ is first bilinearly interpolated to the spatial resolution of the corresponding SAM3 feature map. For each spatial position (i, j), classwise semantic weights are obtained using a temperature-scaled softmax

$$w_c(i,j) = \frac{\exp\left(\mathcal{S}_{\text{sim}}(i,j,c)/\tau\right)}{\sum_{k=1}^{C} \exp\left(\mathcal{S}_{\text{sim}}(i,j,k)/\tau\right)} \tag{3}$$

where τ denotes the temperature.

To distinguish heterogeneous input modalities, each modality $m \in \{\text{rgb}, \text{dsm}\}$ is associated with a learnable embedding

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$e_m \in \mathbb{R}^{d_e}$. A lightweight projection produces a channelwise modality gate

$$g^{(m)} = 2\sigma\left(W_g e_m\right), \quad g^{(m)} \in \mathbb{R}^r. \tag{4}$$

The final LCAS modulation parameters are then obtained by jointly mixing the semantic bases according to spatial semantic confidence and modality identity

$$\tilde{s}^{(l,m)}(i,j) = \sum_{c=1}^{C} w_c(i,j)\left(g^{(m)} \odot s_c^{(l)}\right)$$
$$\tilde{h}^{(l,m)}(i,j) = \sum_{c=1}^{C} w_c(i,j)\left(g^{(m)} \odot h_c^{(l)}\right). \tag{5}$$

W_g and the modality embeddings are initialized such that $g^{(m)} = \mathbf{1}$ for both modalities at the beginning of training. Consequently, LCAS initially behaves identically for RGB and DSM and gradually learns modality-specific modulation during optimization.

Because $\mathcal{S}_{\text{sim}}$ is obtained at the coarse patch resolution of the VLP encoder (e.g., 7×7 for ViT-B/32), LCAS should be interpreted as providing region-level semantic conditioning rather than pixel-accurate boundary guidance. Fine-grained spatial localization is instead recovered through dense supervision and decoder-level prediction (Section III-C).

With this conditioning strategy, modality-specific features are computed as

$$F_{\text{rgb}} = \mathcal{E}\left(I_{\text{rgb}}; \theta_{\text{frozen}}, \Phi_{\text{rgb}}, \mathcal{F}(\mathcal{S}_{\text{sim}}, 0)\right)$$
$$F_{\text{dsm}} = \mathcal{E}\left(I_{\text{dsm}}; \theta_{\text{frozen}}, \Phi_{\text{dsm}}, \mathcal{F}(\mathcal{S}_{\text{sim}}, 1)\right) \tag{6}$$

where Φ_{rgb} and Φ_{dsm} are two independent sets of bottleneck parameters $\{W_{\text{down}}, W_{\text{up}}\}$ per transformer block (4.26 M each). In contrast, the LCAS module $\mathcal{F}(\cdot, m)$ is a single shared conditioner across both modalities: it maintains one shared set of per-class scale and shift bases $\{s_c^{(l)}, h_c^{(l)}\}$, together with the modality embeddings $\{e_m\}$ and the projection W_g that gates these bases. With $r = 32$ and four conditioned blocks, the resulting overhead is approximately 28 K trainable parameters in total. Modality differentiation at the conditioning level is therefore realized by $g^{(m)}$ rather than by duplicating the full conditioner. This formulation decouples semantic, spatial, and modality factors within a unified conditioning framework, enabling fine-grained and interpretable adaptation across heterogeneous multimodal inputs. Because $w_c(i,j)$ is computed from the input-dependent similarity map $\mathcal{S}_{\text{sim}}$, the effective affine modulation in (1) varies across images and spatial positions. Together with the modality gate $g^{(m)}$, LCAS therefore produces image-, position-, and modality-dependent adaptation while retaining a shared frozen backbone. This differs from conventional static adapter parameterizations, whose learned parameters are not explicitly conditioned on spatially resolved external semantic priors. It also differs from globally conditioned FiLM-style modulation [30], which applies one shared affine transformation across spatial positions.

The two insertions deliberately use different coverage. The LMA bottlenecks Φ_{rgb} and Φ_{dsm} are present in all $L = 32$ blocks of the ViT-L+ encoder, whereas LCAS conditioning is applied only at the four global-attention blocks $\mathcal{L}_{\text{LCAS}} = \{7, 15, 23, 31\}$ (zero-based indices) of the ViTDet hybrid-attention schedule [23]. The remaining 28 window-attention blocks use unconditioned adapters, corresponding to $\tilde{s}^{(l,m)}(i,j) = \mathbf{1}$ and $\tilde{h}^{(l,m)}(i,j) = \mathbf{0}$.

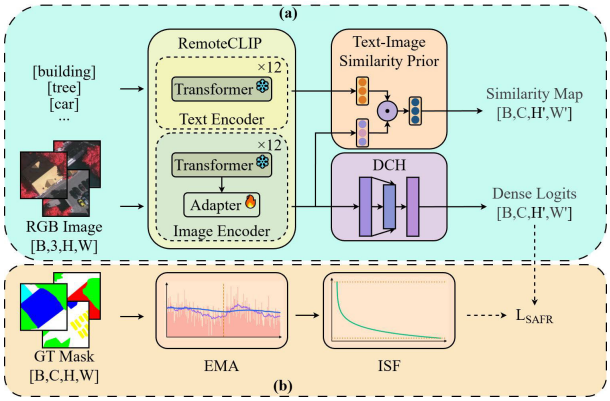


Fig. 4. Detailed architecture of the two core modules. (a) VLP module: a frozen RemoteCLIP encoder with lightweight adapters produces a patch-text similarity map and dense classification logits through two complementary branches. (b) SAFR: ground-truth masks are used to compute patch-level class frequencies, which are temporally smoothed via EMA and converted to ISF weights for loss reweighting.

This sparse placement is motivated by two considerations. First, the vision–language prior provides coarse image-level semantic context, making the global-attention blocks natural locations for integrating such information with features that already aggregate long-range context. Second, the LCAS parameter count grows linearly with the number of conditioned blocks. Conditioning only four blocks therefore provides a substantially lower cost alternative to dense conditioning. The ablation study in Section IV-E empirically evaluates this design choice against shallow-, middle-, deep-, and all-block placements

C. VLP Guidance

While LMA (Section III-B) facilitates spatially adaptive and modality-aware feature transformation, it requires semantically meaningful priors to guide the conditioning process. However, existing vision–language models, despite their rich semantic knowledge, are not directly suitable for dense prediction due to their limited spatial precision. To bridge this gap, we introduce a VLP module that extracts complementary semantic and spatial cues and integrates them into the segmentation pipeline, as schematically illustrated in Fig. 4(a).

We adopt RemoteCLIP [12] as the VLP backbone, leveraging its domain-aligned pretraining on remote sensing image–text pairs. To preserve pretrained representations, the encoder parameters are kept frozen, while lightweight bottleneck adapters following the same residual design as (1) (but without spatial or modality conditioning) are inserted into each transformer block for task-specific adaptation.

To obtain robust class-level semantics, we employ prompt ensembling. For each class c, $K = 6$ hand-crafted templates $\{p_k(c)\}_{k=1}^{K}$ are encoded by the frozen text encoder $\mathcal{T}$, ℓ_2-normalized, averaged, and renormalized to produce the class embedding

$$f_c^t = \text{Normalize}\left(\frac{1}{K}\sum_{k=1}^{K}\frac{\mathcal{T}(p_k(c))}{\|\mathcal{T}(p_k(c))\|}\right). \tag{7}$$

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The templates are "a top-down remote sensing image of {class}," "an aerial view of {class}," "a remote sensing image of {class}," "a top-down view of {class}," "an overhead image of {class}," and "a nadir view of {class}."

Let $\{z_i\}$ denote the visual patch tokens output by the adapted RemoteCLIP image encoder. Given $\{z_i\}$ and class embeddings $\{f_c^t\}$, we derive two complementary priors.

1) Similarity Prior: A patch–text similarity map is computed via cosine similarity

$$\mathcal{S}_{\text{sim}}(i,c) = \frac{\langle \tilde{f}_i^v, f_c^t \rangle}{\|\tilde{f}_i^v\| \cdot \|f_c^t\|}, \quad \tilde{f}_i^v = W_{\text{proj}} \cdot \text{LN}(z_i) \tag{8}$$

where W_{proj} is the frozen visual projection inherited from RemoteCLIP, which maps the hidden dimension to the shared vision-language embedding space. The raw similarity map is then rescaled per-sample via min-max normalization

$$\tilde{\mathcal{S}}_{\text{sim}}(i,c) = \frac{\mathcal{S}_{\text{sim}}(i,c) - \min_{j,k} \mathcal{S}_{\text{sim}}(j,k)}{\max_{j,k} \mathcal{S}_{\text{sim}}(j,k) - \min_{j,k} \mathcal{S}_{\text{sim}}(j,k) + \epsilon}. \tag{9}$$

In the remainder of this article, $\mathcal{S}_{\text{sim}}$ refers to the normalized map $\tilde{\mathcal{S}}_{\text{sim}}$, which serves a dual role: it provides the spatial conditioning signal for LCAS (5) and acts as one of the two priors fused with the decoder output (11). This branch preserves the class-level semantic alignment inherited from large-scale pretraining.

This dual role also constrains the choice of normalization and rules out the two standard alternatives. A softmax across classes is redundant for the first role, since LCAS already applies its own temperature-scaled softmax in (3), and undesirable for the second: enforcing $\sum_c \tilde{\mathcal{S}}_{\text{sim}}(i,c) = 1$ discards the absolute strength of the vision–language match, so a confidently matched patch and an uncertain one would contribute additive terms of equal magnitude, preventing the gate $\sigma(w_{\text{sim}})$ from learning to trust the prior selectively. Per-patch ℓ_2 normalization is scale-invariant in the same sense and additionally leaves the map unbounded below, so negative-signed evidence can enter (11) before γ has learned to attenuate it. Min–max preserves the confidence ordering while bounding the map to $[0, 1]$, which is also what makes the scale reconciliation discussed after (11) well posed. We normalize per sample rather than per batch because the attainable similarity range of a contrastive encoder is image-dependent: a batch-level statistic would make each sample's conditioning strength depend on the other images in its minibatch.

2) Dense Prior: To enhance spatial precision, we introduce a DCH that operates on the same adapted patch tokens $\{z_i\}$. Each token is first refined through a residual MLP block and then linearly projected to the number of classes

$$\hat{z}_i = z_i + \text{MLP}(\text{LN}(z_i)), \quad \mathcal{Y}_{\text{dense}}(i) = W_{\text{cls}} \cdot \text{LN}(\hat{z}_i) \tag{10}$$

where MLP consists of two linear layers with a Gaussian error linear unit (GELU) activation and an expansion ratio of $4\times$, and $W_{\text{cls}} \in \mathbb{R}^{C \times d_h}$ is a learnable classification projection. The resulting logits $\mathcal{Y}_{\text{dense}} \in \mathbb{R}^{B \times C \times H' \times W'}$ are supervised using ground-truth masks downsampled to the patch resolution.

The two priors are adaptively integrated with the decoder output

$$Y = \hat{Y} + \sigma(\gamma) \cdot \left(\sigma(w_{\text{sim}}) \odot \mathcal{S}_{\text{sim}}^{\uparrow} + \sigma(w_{\text{cls}}) \odot \mathcal{Y}_{\text{dense}}^{\uparrow} \right) \tag{11}$$

where $w_{\text{sim}}, w_{\text{cls}} \in \mathbb{R}^C$ are learnable per-class weights broadcast along the spatial dimensions, γ is a scalar that controls the overall contribution of VLP, and $\uparrow$ denotes bilinear upsampling to the output resolution. The parameter γ is initialized to zero for stable training. Note that $\mathcal{S}_{\text{sim}}$ is bounded in $[0, 1]$ after the normalization in (9), whereas $\mathcal{Y}_{\text{dense}}$ and $\hat{Y}$ are unnormalized logits; the per-class gates $\sigma(w_{\text{sim}})$, $\sigma(w_{\text{cls}})$ and the overall gate $\sigma(\gamma)$ provide learnable scale reconciliation between the two sources.

The final prediction Y is optimized using the standard cross-entropy loss $\mathcal{L}_{\text{seg}}$.

Overall, the VLP module provides complementary semantic and spatial priors: the similarity branch supplies coarse but semantically rich guidance that conditions both the LCAS adaptation (Section III-B) and the decoder-level fusion, while the DCH branch enhances spatial precision through dense supervision.

D. SAFR Loss

The DCH (Section III-C) necessitates patch-level supervision derived from ground-truth annotations; however, multimodal remote sensing datasets are frequently characterized by acute class imbalances. Dominant categories typically occupy vast spatial extents, whereas semantically critical classes exhibit sparse occurrences. Under such skewed distributions, the standard cross-entropy objective becomes disproportionately influenced by high-frequency categories, precipitating biased optimization trajectories and suboptimal recognition of minority classes.

While static class-balancing strategies based on global dataset statistics partially alleviate this issue, they fail to capture the stochastic nature of training, where class distributions fluctuate across minibatches, and feature representations evolve over time. To address this limitation, we draw on the principle of inverse frequency weighting [26] and adapt its core idea to the dense prediction setting. In particular, we develop the SAFR mechanism, which performs online class reweighting with temporal smoothing to dynamically adjust per-class loss contributions based on their spatial coverage in each minibatch, as illustrated in Fig. 4(b).

For a training batch at iteration t, let N denote the total number of patches and $y_i \in \{1, \ldots, C\}$ the ground-truth label of patch i. The empirical spatial frequency of class c, defined as the fraction of patches occupied by that class, is

$$\hat{f}_c^{(t)} = \frac{|\{i \mid y_i = c\}|}{N}. \tag{12}$$

To obtain stable estimates over time, we maintain an EMA

$$f_c^{(t)} = \beta_1 f_c^{(t-1)} + (1 - \beta_1)\, \hat{f}_c^{(t)} \tag{13}$$

where $\beta_1 \in [0, 1)$ is a momentum coefficient and $f_c^{(0)} = \hat{f}_c^{(0)}$. This update captures both long-term dataset statistics and short-term batch fluctuations.

Based on the smoothed frequency $f_c^{(t)}$, we define the inverse spatial frequency (ISF) weight as

$$w_c^{ISF} = \log \frac{1 + \epsilon}{f_c^{(t)} + \epsilon} \tag{14}$$

where ϵ is a small constant for numerical stability. This formulation assigns larger weights to classes with low spatial

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coverage and smaller weights to dominant ones, effectively capturing class imbalance as measured by per-class spatial coverage (i.e., the fraction of patches occupied by each class within each minibatch), rather than relying on fixed global dataset statistics. The resulting weight w_c^{ISF} is shared across all spatial positions of the same class, as the estimation operates at the class level over dense patch grids.

To avoid extreme values that may destabilize optimization, the ISF weights are linearly normalized to a bounded interval [0.5, 1.5]

$$\tilde{w}_c^{ISF} = 0.5 + \frac{w_c^{ISF} - \min_k w_k^{ISF}}{\max_k w_k^{ISF} - \min_k w_k^{ISF} + \epsilon}. \tag{15}$$

To decouple frequency estimation from direct loss modulation and prevent transient batch noise from propagating into the optimization process, we further apply a second-stage EMA to the normalized weights. The final class weight is defined as

$$w_c^{(t)} = \begin{cases} 1, & t < T_{\text{warm}} \\ \beta_2 w_c^{(t-1)} + (1-\beta_2)\,\tilde{w}_c^{ISF}, & t \geq T_{\text{warm}} \end{cases} \tag{16}$$

where $\beta_2 \in [0, 1)$ is a second momentum coefficient and T_{warm} denotes the warmup duration. During warmup, uniform weights are used to ensure stable early optimization; thereafter, adaptive reweighting is gradually introduced through temporally smoothed updates. Unless otherwise stated, we set $\beta_1 = 0.99$, $\beta_2 = 0.9$, $T_{\text{warm}} = 100$ iterations, and $\epsilon = 10^{-6}$. Fig. 5 illustrates the weight dynamics on Vaihingen: the weights rapidly differentiate after warmup and stabilize within approximately 5000 iterations, with rare classes (car) receiving significantly higher weights than dominant ones (imp. surf.).

Let $\mathbf{z}^{(i)} \in \mathbb{R}^C$ denote the logits predicted by the DCH for patch i, and let $y_i^{\downarrow}$ be the ground-truth label downsampled to the patch resolution. The SAFR auxiliary loss applies standard cross-entropy at each patch position, weighted by the adaptive class weight $w_{y_i^{\downarrow}}^{(t)}$ corresponding to its ground-truth label

$$\mathcal{L}_{SAFR} = \frac{\sum_{i=1}^{N} w_{y_i^{\downarrow}}^{(t)} \cdot \text{CE}\left(\mathbf{z}^{(i)}, y_i^{\downarrow}\right)}{\sum_{i=1}^{N} w_{y_i^{\downarrow}}^{(t)}}. \tag{17}$$

The weighted-mean formulation preserves gradient-scale stability while dynamically reallocating learning emphasis toward underrepresented categories.

The final training objective combines the primary segmentation loss and the SAFR auxiliary term.

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \lambda \mathcal{L}_{SAFR} \tag{18}$$

where λ controls the contribution of the adaptive reweighting objective (set to 0.1 by default). Unlike static reweighting strategies, SAFR models online class coverage dynamics, providing coverage-aware weighting for the patch-level dense branch introduced in Section III-C. We emphasize that SAFR is designed specifically to supervise this branch rather than as an architecture-agnostic class-balancing loss: its statistics are estimated over the N patch positions of the dense head while extending the same formulation to arbitrary segmentation architectures operating at pixel resolution is left for future work.

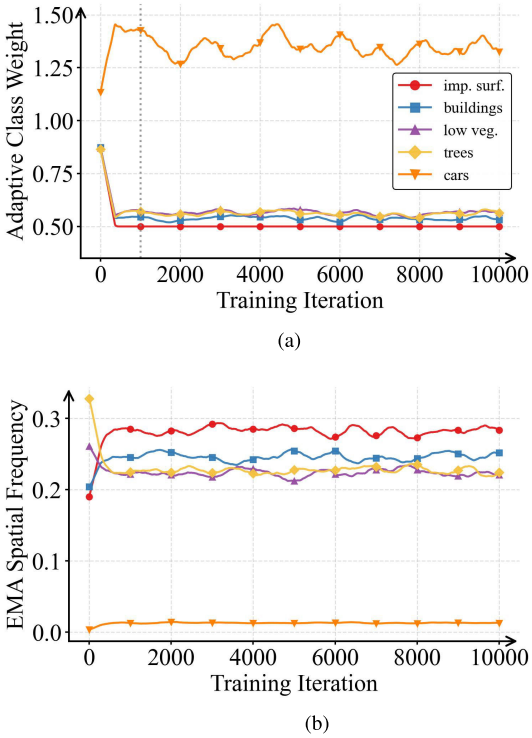


Fig. 5. SAFR weight dynamics during training on Vaihingen. (a) Adaptive class weights: the car class receives the highest weight (~1.35), while dominant classes such as imp. surf. are suppressed to the lower bound of 0.5. The residual fluctuations in the car weight reflect its sparse and uneven distribution across training patches, demonstrating that SAFR dynamically tracks batch-level class coverage rather than relying on fixed dataset statistics. (b) EMA-smoothed class frequency estimates converge rapidly after the initial iterations.

IV. Empirical Results

A. Datasets

1) ISPRS Vaihingen: The International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen benchmark consists of 33 aerial image tiles acquired over a residential area in Germany, with a ground sampling distance (GSD) of 9 cm. Each tile includes three spectral bands (near-infrared, red, and green) along with a DSM generated via dense image matching. Following the standard protocol [31], 12 tiles are used for training and 4 for testing. The dataset provides pixelwise annotations for six classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter.

2) ISPRS Potsdam: The ISPRS Potsdam dataset comprises 38 high-resolution aerial image tiles with a GSD of 5 cm, capturing complex urban scenes with dense building structures. Each tile contains four spectral channels (R, G, B, and NIR) and a normalized DSM. Following standard evaluation settings, 18 tiles are used for training and six for testing. The annotation scheme is consistent with the Vaihingen dataset, including the same six semantic classes.

3) MMHunan: The MMHunan dataset [32] is a large-scale land cover dataset covering diverse regions in Hunan Province, China, with a spatial resolution of 10 m. It consists of 500 Sentinel-2 image patches of size 256 × 256, each paired with SRTM-derived elevation data. In this work, the RGB channels are used as the optical modality. Following [11], 450 patches are used for training and 50 for validation. The

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dataset includes seven land cover categories: cropland, forest, grassland, wetland, water, unused land, and built-up area.

B. Implementation Details

1) Architectural Configuration: The MAPLE framework adopts the ViTDet [23] visual encoder from the SAM3 [5] PE [6] as the backbone, specifically the ViT-L+ variant. The backbone is kept frozen during training to preserve pretrained feature representations. In addition, the VLP is introduced via a frozen RemoteCLIP [12] encoder based on the ViT-B/32 architecture. The decoder follows the UNetFormer design [29], comprising two global–local attention blocks, two weighted-fusion modules that progressively merge multiscale features, and a feature-refinement head.

2) Optimization Protocol: All experiments are implemented in PyTorch and conducted on a single NVIDIA GeForce RTX 2080 Ti GPU 11 GB video random access memory (11 GB VRAM). The model is optimized using stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 5×10^{-4}. The initial learning rate is set to 0.01 and decayed by a factor of 0.1 at epochs 25, 35, and 45. Training is performed for 50 epochs with a batch size of 4.

3) Data Augmentation and Inference: Training samples are 256×256 crops extracted at uniformly random positions from randomly selected training tiles, with 1000 randomly sampled crops per epoch for the ISPRS benchmarks. For MMHunan, 500 patches are sampled per epoch; since all patches are already 256×256, no additional cropping is applied. Optical images are scaled to $[0, 1]$, while each DSM tile is independently min–max normalized to $[0, 1]$. Data augmentation consists of random vertical and horizontal flips, each applied with probability 0.5; their composition also realizes a 180° rotation. Both transformations are sampled once per training sample and applied identically to the optical image, DSM, and label, thereby preserving intermodality registration. During inference, a sliding-window approach is adopted with dataset-specific strides of 32 for Vaihingen, 128 for Potsdam, and 256 for MMHunan. Logits from overlapping windows are accumulated by summation before applying the arg-max operation. All methods are trained and evaluated under the same protocol, including identical augmentation, crop size, train/test tile split, and sliding-window inference.

C. Evaluation Metrics

Semantic segmentation performance is evaluated using three standard metrics: overall accuracy (OA), mean $F1$-score (mF1), and mean intersection over union (mIoU). OA measures the overall pixel-wise classification accuracy across the entire dataset. In contrast, mF1 and mIoU provide class-level evaluation, where mF1 computes the harmonic mean of precision and recall, and mIoU measures the overlap between predicted and ground-truth regions.

Following the ISPRS evaluation protocol in [11], the "clutter" class is excluded from the computation of mF1 and mIoU, ensuring that the evaluation focuses on the remaining semantic categories. This combination of metrics provides a comprehensive assessment of both classification performance and segmentation quality.

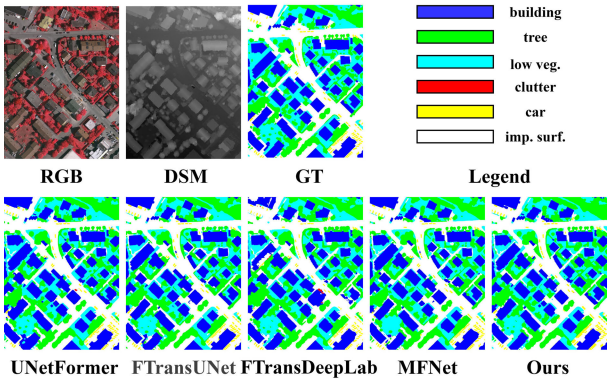


Fig. 6. Qualitative comparison on the ISPRS Vaihingen dataset. MAPLE produces more accurate segmentation for small objects (e.g., cars) and clearer boundaries between low vegetation and trees compared with competing methods.

D. Main Results

We benchmark MAPLE against 16 state-of-the-art methods, including PSPNet [33], MAResU-Net [34], V-FuseNet [31], FuseNet [36], ESANet [37], SA-Gate [18], CMGFNet [38], TransUNet [40], CMFNet [1], UNetFormer [29], MFTransNet [39], FTransUNet [2], RS3Mamba [35], FTransDeepLab [41], MultiSenseSeg [42], and MFNet [11]. Most of these methods are tailored for remote sensing segmentation tasks. In our experiments, PSPNet, MAResU-Net, UNetFormer, and RS3Mamba are evaluated using only optical inputs to isolate the contribution of DSM data. This setup enables a direct comparison between single-modal and multimodal approaches, thereby quantifying the performance gains introduced by multimodal fusion.

1) Results on ISPRS Vaihingen: Quantitative results on the ISPRS Vaihingen benchmark are reported in Table I. MAPLE achieves state-of-the-art performance, with an mF1 score of 92.52% and an mIoU of 86.37%, outperforming the second-best method, MFNet, by 0.81% and 1.34%, respectively. A clear performance gap is observed between single-modal and multimodal approaches: single-modal methods achieve mIoU scores in the range of 76.96%–82.78%, while most multimodal methods exceed 81%, indicating the benefit of incorporating geometric information from DSM. At the class level, MAPLE shows notable improvements in the "Low vegetation" and "Car" categories, with gains of 4.93% and 4.39% over the next-best results. These improvements suggest that integrating vision-language semantic guidance with dense supervision effectively mitigates spectral ambiguities between visually similar classes. In particular, the gains in spatially ambiguous categories can be further attributed to the proposed LCAS mechanism, which enables position-aware feature modulation guided by CLIP-derived semantic priors. On the other hand, MAPLE does not uniformly dominate every category: MFNet [11] retains the best per-class scores on *Building* and *Tree*, with gaps within 2.71 percentage points. MAPLE nonetheless remains the top method in terms of the aggregated mF1 and mIoU, and achieves this at a fraction of MFNet's trainable parameter budget.

Qualitative comparisons in Fig. 6 further support these observations. In the first scenario, competing methods often misclassify low vegetation as trees in shadowed regions,

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TABLE I

QUANTITATIVE RESULTS ON THE ISPRS VAIHINGEN DATASET. THE BEST RESULTS ARE IN BOLD. THE SECOND-BEST RESULTS ARE UNDERLINED (%)

Method	Backbone	OA						mF1	mIoU
		Bui.	Tre.	Low.	Car	Imp.	Total		
PSPNet [33]	ResNet101	94.52	90.17	78.84	79.22	92.03	89.94	86.55	76.96
MAResU-Net [34]	ResNet18	94.84	89.99	79.09	85.89	92.19	90.17	88.54	79.89
UNetFormer [29]	ResNet18	96.23	91.85	79.95	86.99	91.85	91.17	89.48	81.97
RS^{3}Mamba [35]	R18-Mamba-T	97.40	92.14	79.56	88.15	92.19	91.64	90.34	82.78
FuseNet [36]	VGG16	96.28	90.28	78.98	81.37	91.66	90.51	87.71	78.71
V-FuseNet [31]	VGG16	95.92	91.36	77.64	76.06	91.85	90.49	87.89	78.92
ESANet [37]	ResNet34	95.69	90.50	77.16	85.46	91.39	90.61	88.18	79.42
SA-Gate [18]	ResNet101	94.84	92.56	81.29	87.79	91.69	91.10	89.81	81.27
CMGFNet [38]	ResNet34	97.75	91.60	80.03	87.28	92.35	91.72	90.00	82.26
CMFNet [1]	VGG16	97.17	90.82	80.37	85.47	92.36	91.40	89.48	81.44
MFTransNet [39]	ResNet34	96.41	91.48	80.09	86.52	92.11	91.22	89.62	81.61
TransUNet [40]	R50-ViT-B	96.48	92.77	76.14	69.56	91.96	90.96	87.34	78.26
FTransUNet [2]	R50-ViT-B	98.20	91.94	81.49	91.27	93.01	92.40	91.21	84.23
FTransDeepLab [41]	ResNet101	98.11	93.45	80.35	89.98	93.23	92.61	91.00	83.87
MultiSenseSeg [42]	Segformer-B2	97.91	93.04	81.58	89.06	93.56	92.73	91.42	84.53
MFNet [11]	ViT-H	**98.38**	**93.94**	80.70	90.47	**93.59**	92.97	91.71	85.03
MAPLE (Ours)	ViT-L+	97.49	91.23	**86.51**	**95.66**	93.58	**93.00**	**92.52**	**86.37**

TABLE II

QUANTITATIVE RESULTS ON THE ISPRS POTSDAM DATASET. THE BEST RESULTS ARE IN BOLD. THE SECOND-BEST RESULTS ARE UNDERLINED (%)

Method	Backbone	OA						mF1	mIoU
		Bui.	Tre.	Low.	Car	Imp.	Total		
PSPNet [33]	ResNet101	97.03	83.13	85.67	88.81	90.91	88.67	88.92	80.36
MAResU-Net [34]	ResNet18	96.82	83.97	87.70	95.88	92.19	89.82	90.86	83.61
UNetFormer [29]	ResNet18	97.69	86.47	87.93	95.91	92.27	90.65	91.71	85.05
RS^{3}Mamba [35]	R18-Mamba-T	97.70	86.11	89.53	96.23	91.36	90.49	91.69	85.01
FuseNet [36]	VGG16	97.48	85.14	87.31	96.10	92.64	90.58	91.60	84.86
V-FuseNet [31]	VGG16	97.23	84.29	89.03	95.49	91.62	90.22	91.26	84.26
ESANet [37]	ResNet34	97.10	85.31	87.81	94.08	92.76	89.74	91.22	84.15
SA-Gate [18]	ResNet101	96.54	81.18	85.35	96.63	90.77	87.91	90.26	82.53
CMGFNet [38]	ResNet34	97.41	86.80	86.68	95.68	92.60	90.21	91.40	84.53
CMFNet [1]	VGG16	97.63	87.40	88.00	95.68	92.84	91.16	92.10	85.63
MFTransNet [39]	ResNet34	97.37	85.71	86.92	96.05	92.45	89.96	91.11	84.04
TransUNet [40]	R50-ViT-B	96.63	82.65	89.98	93.17	91.93	90.01	90.97	83.74
FTransUNet [2]	R50-ViT-B	97.78	**88.27**	88.48	96.31	93.17	91.34	92.41	86.20
FTransDeepLab [41]	ResNet101	97.58	85.87	90.08	96.94	92.81	90.97	92.08	85.62
MultiSenseSeg [42]	Segformer-B2	98.32	87.65	89.54	96.27	92.46	91.30	92.35	86.10
MFNet [11]	ViT-H	**98.44**	87.37	**90.36**	96.24	93.17	91.71	92.70	86.69
MAPLE (Ours)	ViT-L+	98.40	87.43	89.02	**97.26**	**94.31**	**91.86**	**92.89**	**87.01**

whereas MAPLE produces more accurate predictions. In the second scenario, many methods fail to detect small cars or produce fragmented masks, while MAPLE recovers more complete objects with clearer boundaries. This improvement can be attributed in part to the SAFR loss, which adaptively enhances supervision for spatially sparse categories during training.

2) Results on ISPRS Potsdam: Quantitative results on the ISPRS Potsdam benchmark are reported in Table II. MAPLE achieves the best performance across all evaluation metrics, with a Total OA of 91.86%, an mF1 of 92.89%, and an mIoU of 87.01%. Compared to Vaihingen, Potsdam presents a more challenging setting due to its higher spatial resolution (5 cm) and more complex urban structures, requiring finer boundary delineation and improved intra-class consistency. MAPLE shows notable improvements in the "Car" and "Impervious surfaces" categories, achieving accuracies of 97.26% and 94.31%, respectively. The latter category exhibits substantial appearance variability across different functional regions such as roads, sidewalks, and parking lots. By leveraging pretrained vision–language representations together with parameter-efficient adaptation, MAPLE

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TABLE III
QUANTITATIVE RESULTS (%) ON THE MMHUNAN DATASET

Method	Backbone	OA								mF1	mIoU
		Cropland	Forest	Grassland	Wetland	Water	Unused	Built-up	Total		
FuseNet [36]	VGG16	76.10	89.21	37.11	8.37	70.12	72.75	16.99	76.35	54.80	42.30
CMFNet [1]	VGG16	**83.83**	82.27	**43.88**	**39.57**	70.14	**81.72**	23.66	77.41	59.63	46.52
MFTransNet [39]	ResNet34	78.99	**91.30**	33.80	26.80	77.38	78.29	27.17	80.07	61.45	48.25
FTransUNet [2]	R50-ViT-B	78.75	90.54	32.13	27.51	76.64	75.59	**48.63**	79.47	61.95	48.78
CMGFNet [38]	ResNet34	82.08	87.90	37.50	26.48	79.82	74.64	41.15	79.85	62.69	49.44
FTransDeepLab [41]	ResNet101	79.39	88.89	35.71	30.88	83.95	78.14	32.60	80.62	62.51	49.66
MultiSenseSeg [42]	Segformer-B2	78.03	90.93	40.47	38.16	80.19	81.03	38.14	80.51	63.74	50.76
MFNet [11]	ViT-H	79.66	90.06	42.61	38.81	80.31	78.92	40.04	81.07	64.13	51.08
MAPLE (Ours)	ViT-L+	79.26	90.25	35.87	32.49	**90.60**	77.77	37.97	**82.37**	**65.45**	**52.42**

TABLE IV
FOUR CATEGORIES ON WHICH MAPLE TRAILS THE STRONGEST BASELINE. VALUES ARE PER-CLASS OA (RECALL), FOLLOWING THE CONVENTION OF TABLE III. "PRED./TRUE" IS MAPLE'S RATIO OF PREDICTED TO ACTUAL PIXEL COUNTS; A VALUE BELOW ONE INDICATES THAT THE CATEGORY IS PREDICTED LESS OFTEN THAN IT OCCURS

Class	MAPLE OA	Best other OA	Δ	Pred./true
Cropland	79.26	83.83	−4.57	0.98
Grassland	35.87	43.88	−8.01	0.76
Wetland	32.49	39.57	−7.08	0.77
Built-up area	37.97	48.63	−10.66	0.67

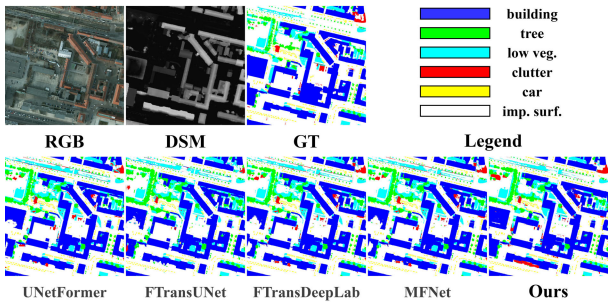


Fig. 7. Qualitative comparison on the ISPRS Potsdam dataset. MAPLE better preserves the structural integrity of buildings and impervious surfaces in complex urban layouts.

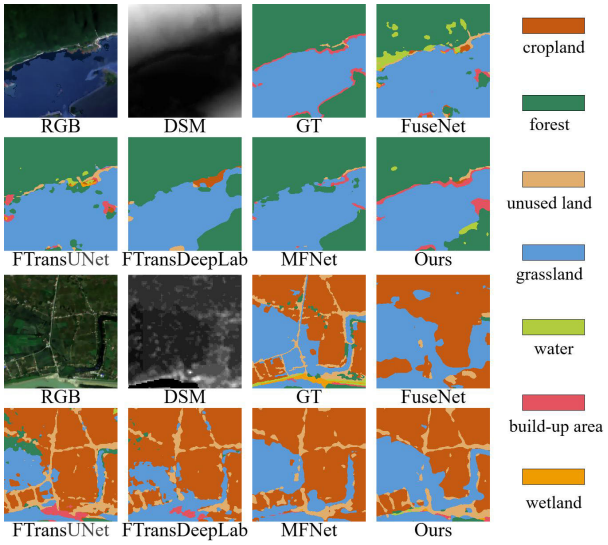


Fig. 8. Qualitative comparison on the MMHunan dataset. Two representative scenes are shown. MAPLE achieves more coherent water and forest segmentation, though some confusion between cropland and grassland persists at 10 m resolution.

effectively captures these variations while benefiting from RGB–DSM complementarity. The relative mIoU improvement over the second-best method (MFNet) is smaller on Potsdam (0.32%) than on Vaihingen (1.34%). This difference can be attributed in part to the higher spatial resolution of Potsdam, which provides richer visual details and reduces the relative contribution of VLPs. Nevertheless, MAPLE attains slightly higher segmentation accuracy than MFNet while using substantially fewer task-specific trainable parameters. In particular, MAPLE uses 19.51 M trainable parameters compared with 111.28 M for MFNet, corresponding to an approximately 82.5% reduction in task-specific trainable parameters.

Qualitative results in Fig. 7 further support these findings. In the first case, FTransUNet and MFNet produce irregular building boundaries near roof edges, whereas MAPLE generates more regular and coherent outlines, benefiting from the spatial modeling of the DCH branch. In the second case, many baselines struggle to distinguish impervious surfaces from adjacent low vegetation, often resulting in boundary inconsistencies. On the contrary, MAPLE produces more accurate transitions between classes, reflecting the effectiveness of multimodal fusion and vision–language guidance.

3) Results on MMHunan: Table III presents quantitative results on the MMHunan dataset, which has a coarser spatial resolution (10 m) and severe class imbalance. MAPLE achieves the best performance overall, with a Total OA of 82.37%, an mF1 of 65.45%, and an mIoU of 52.42%, outperforming the second-best method, MFNet, by 1.30%, 1.32%, and 1.34%, respectively. Notably, MAPLE shows a substantial improvement in the "Water" category, reaching an accuracy of 90.60%, exceeding FTransDeepLab by 6.65%. This result is consistent with the intended role of SAFR under class imbalance, although the gains are not uniform across all difficult categories. Qualitative results in Fig. 8 further illustrate that MAPLE produces more spatially coherent predictions for water and forest regions compared to baseline methods.

The per-class values in Table III correspond to recall and therefore reflect both class separability and the model's prediction bias. Table IV isolates the four categories on which MAPLE trails the strongest competing result. Three of them are under-predicted (pred./true 0.76, 0.77, and 0.67), with systematic confusion patterns: 41.7% of grassland is assigned to forest, 43.2% of wetland to water, and 24.8% of built-up

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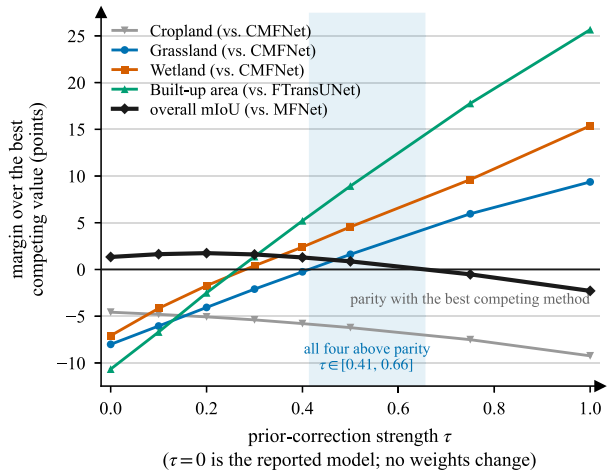


Fig. 9. MAPLE's operating curve on MMHunan. The decision rule is shifted along the class-prior direction, $\hat{y} = \arg\max_c(z_c - \tau \log \pi_c)$, with π the training-set class frequency; the network weights are unchanged and $\tau = 0$ is the model reported in Table III. Each curve is the margin of MAPLE over the best competing value for that quantity, so zero is parity. Within the shaded interval, MAPLE simultaneously exceeds the best reported per-class OA on grassland, wetland, and built-up area while retaining the best overall mIoU. Cropland, a frequent category, is penalized rather than helped by the same correction.

area to unused land. The comparison also reveals a class-dependent tradeoff: CMFNet gains 8.01 percentage points on grassland relative to MAPLE, but performs 20.46 and 7.98 points worse on water and forest, respectively, with an overall mIoU 5.90 points lower. Built-up additionally favors FTransUNet, confirming that MAPLE's overall advantage is not uniform across categories.

To examine whether these lower recalls are partly associated with the default decision operating point, we apply posthoc logit adjustment along the class-prior direction (Fig. 9)

$$\hat{y} = \arg\max_c \left(z_c - \tau \log \pi_c\right) \tag{19}$$

where π_c denotes the training-set frequency of class c and z_c is the corresponding logit of the unchanged network. At $\tau = 0.5$, the recalls of grassland, wetland, and built-up increase to 45.50%, 44.13%, and 57.56%, respectively, exceeding the strongest competing values of 43.88%, 39.57%, and 48.63%, while MAPLE retains 51.95% mIoU versus 51.08% for the strongest baseline. These results suggest that much of the deficit on these three categories is associated with the default class-decision bias rather than a complete lack of separability. Cropland behaves differently, with recall decreasing as τ increases, consistent with its near-balanced prediction-to-ground-truth ratio of 0.98. We retain $\tau = 0$ for all primary comparisons to avoid posthoc calibration. Overall, SAFR improves robustness to class imbalance rather than guaranteeing gains for every difficult category, while vision–language guidance does not fully eliminate ambiguity among spectrally similar classes.

E. Ablation Study

To assess the contribution of each component, we conduct ablations on the Vaihingen dataset (Table V). Rows are grouped by the mechanism they interrogate, and the discussion below follows the same grouping in the same order: each block reports both the removal of a mechanism and the alternative designs considered in its place, so that a component's contribution and the design choice that realizes it can be read together. Fig. 10 visualizes these effects.

TABLE V

ABLATION STUDY ON MAPLE COMPONENTS ON VAIHINGEN (%). BLOCKS CORRESPOND ONE-TO-ONE WITH THE PARAGRAPHS OF THIS SUBSECTION AND APPEAR IN THE SAME ORDER. IN BLOCK B, "W/O LCAS" LEAVES THE PRIOR ACTING AT THE DECODER ONLY AND "W/O DECODER FUSION" LEAVES IT CONDITIONING THE ADAPTERS ONLY, WHILE "W/O SPATIAL LAYER" REDUCES THE CONDITIONING TO A SINGLE GLOBAL, FILM-STYLE AFFINE TRANSFORMATION PER CHANNEL. IN BLOCK D, THE RGB-ONLY AND DSM-ONLY VARIANTS DISABLE LANGUAGE GUIDANCE TO ISOLATE MODALITY CONTRIBUTIONS, WHEREAS THE LEARNED DSM STEM IS EVALUATED WITHIN THE FULL MAPLE CONFIGURATION TO EXAMINE THE DSM INPUT ENCODING

Configuration	OA	mF1	mIoU
MAPLE (Full)	**93.00**	**92.52**	**86.37**
A. Impact of VLP guidance			
w/o VLP Guidance	91.85	91.16	84.41
w/o Similarity Branch	92.16	91.62	84.93
w/o DCH Branch	92.53	91.95	85.58
B. Impact of LCAS			
w/o LCAS	92.37	91.86	85.17
w/o Decoder fusion	92.45	91.46	84.54
w/o Spatial layer	92.61	92.14	85.73
Random spatial prior	91.91	90.99	83.83
Learned prior, VL-independent	92.78	92.15	85.75
C. Impact of SAFR			
w/o SAFR Loss	92.41	91.71	85.47
w/o Adaptive ISF	92.69	92.13	85.79
D. Input modalities and DSM encoding			
RGB only, no prior	91.95	90.87	83.56
DSM only, no prior	85.14	79.98	68.08
Learned $1 \rightarrow 3$ conv stem for DSM	92.97	92.40	86.16

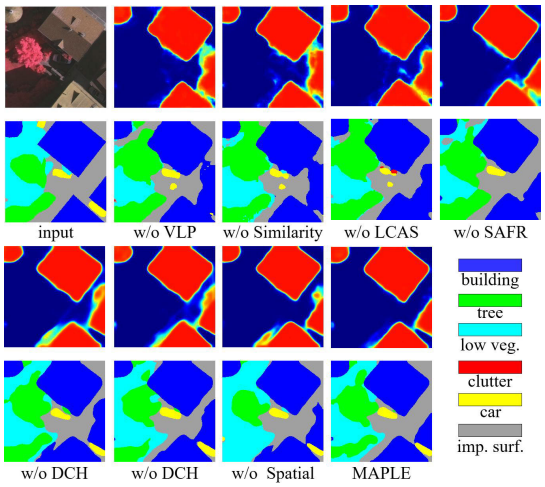


Fig. 10. Visual ablation results on the ISPRS Vaihingen dataset. For each configuration, the top row displays the class activation heatmap for the building category, and the bottom row shows the corresponding segmentation map.

1) Impact of VLP Guidance: Removing the VLP branch simultaneously disables: 1) the decoder-level fusion; and 2) the spatial conditioning signal consumed by LCAS, because both stem from the same RemoteCLIP outputs. The 1.96%

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TABLE VI

ABLATION ON LCAS PLACEMENT (ISPRS VAIHINGEN). $\mathcal{L}_{\text{LCAS}}$ LISTS THE VIT BLOCK INDICES THAT RECEIVE LANGUAGE CONDITIONING; THE LMA BOTTLENECKS ARE PRESENT IN ALL 32 BLOCKS IN EVERY ROW, SO THE ONLY VARIABLE IS WHERE THE CONDITIONING ACTS. THE "NONE" ROW IS THE "W/O LCAS" CONFIGURATION OF TABLE V

Placement	$\mathcal{L}_{\text{LCAS}}$	Depth span	#LCAS params	mIoU (%)
None (unconditioned)	$\varnothing$	—	0	85.17
Shallow	$\{1, 3, 5, 7\}$	early only	28 K	85.66
Middle	$\{11, 13, 15, 17\}$	middle only	28 K	85.89
Deep	$\{25, 27, 29, 31\}$	late only	28 K	85.27
All blocks	$\{0, \ldots, 31\}$	full (dense)	225 K	86.13
Global-attn. (ours)	$\mathbf{\{7, 15, 23, 31\}}$	**full (sparse)**	**28 K**	**86.37**

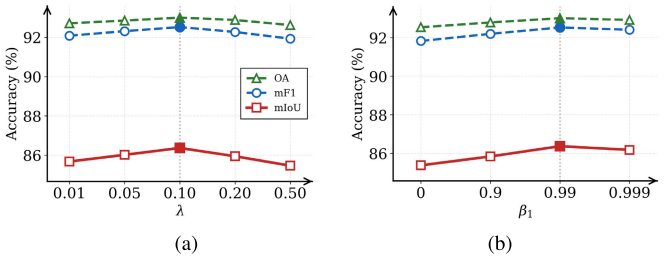


Fig. 11. Sensitivity analysis of SAFR hyperparameters on Vaihingen. (a) Effect of loss weight λ: performance peaks at $\lambda = 0.1$. (b) Effect of EMA momentum β_1: $\beta_1 = 0.99$ achieves the best balance between responsiveness and stability.

mIoU drop reported for "w/o VLP Guidance" is therefore the *joint* contribution of VLPs to decoding and adaptation, and is the largest attributable to any single component. Within the branch, the two priors are complementary but not interchangeable: removing the similarity branch costs 1.44% and removing the DCH branch 0.79%, which is consistent with their intended roles—the former supplies the semantic evidence that additionally conditions LCAS, whereas the latter supplies only the spatial precision that the similarity map lacks.

2) Impact of LCAS: LCAS governs how the prior established above is consumed. Removing it ("w/o LCAS") reduces LMA to content-agnostic adapters and costs 1.20% mIoU, which verifies the effectiveness of language-conditioned modulation. The prior can enter the network at two points, and this block separates them. Measured against the no-prior baseline (84.41%), conditioning the adapters alone contributes + 0.13% and decoder-level fusion alone + 0.76%, yet the two together contribute + 1.96%, exceeding the sum of the individual effects by 1.07%. The two injection points are therefore not independent pathways but halves of a single mechanism, in which the backbone is adapted to produce features that the prior can subsequently correct.

Granularity and content are both essential to that mechanism. Restricting the conditioning to one global affine transformation per channel ("w/o Spatial layer"), which reduces (5) to conventional FiLM-style modulation, costs 0.64%. Conversely, a spatially varying signal is not sufficient on its own: a random spatial prior of identical shape is worse than supplying no prior at all (83.83% versus 84.41%), and a learned prior of the same shape, trained end-to-end but carrying no vision-language information, reaches 85.75%—still 0.62 percentage points below the full RemoteCLIP-conditioned model (86.37%). This gap suggests that the benefit of LCAS is not explained solely by introducing a spatially varying conditioning signal; semantic information inherited from vision–language pretraining provides an additional contribution.

Table VI varies which blocks receive conditioning while keeping the LMA bottlenecks in all 32 blocks, so that placement is the only variable. The "None" row reproduces the "w/o LCAS" configuration and serves as the reference point: conditioning is worth 1.20% mIoU in total. At a matched budget of 28 K conditioning parameters, placement determines how much of that 1.20% is recovered: 0.10% for deep-only, i.e., almost nothing, 0.49% for shallow-only, 0.72% for middle-only, and all of it for the global-attention set $\{7, 15, 23, 31\}$, the only 28 K variant that spans the full depth of the encoder. Conditioning all 32 blocks recovers 0.96 points over the unconditioned baseline and does not outperform the proposed configuration despite using approximately 8× more LCAS parameters. Among the parameter-matched four-block variants evaluated here, the global-attention placement $\{7, 15, 23, 31\}$ provides the strongest result while spanning the full encoder depth. We therefore adopt this sparse placement as the best accuracy–parameter tradeoff among the tested configurations.

3) Impact of SAFR: Removing SAFR ("w/o SAFR Loss") costs 0.90% mIoU, confirming the benefit of auxiliary patch-level supervision applied to the dense branch that VLP introduces. Replacing the adaptive ISF weights with uniform weighting retains that supervision but discards the frequency-aware reweighting, which decomposes the 0.90% into two parts: uniform patch-level supervision accounts for 0.32% and the adaptive reweighting for the remaining 0.58%. The reweighting is thus the larger of the two contributions, and is what distinguishes SAFR from a plain auxiliary loss.

We further note that "w/o Adaptive ISF" constitutes a strong reference point in its own right: at 85.79% mIoU, it already exceeds every published method in Table I, including MFNet (85.03%). Enabling the adaptive reweighting on top of this configuration yields a further + 0.58 percentage points under an otherwise identical architecture, training schedule, and loss weight. Thus, the incremental benefit of adaptive reweighting is measured against an already competitive reference configuration rather than a weak baseline.

Fig. 11(a) shows that performance peaks at $\lambda = 0.1$, with graceful degradation for both smaller and larger values. Fig. 11(b) illustrates the effect of the EMA momentum β_1: without smoothing ($\beta_1 = 0$) performance drops due to unstable estimates, while $\beta_1 = 0.99$ achieves the best balance between stability and adaptability. SAFR is therefore robust within a broad range of hyperparameters.

4) Input Modalities and DSM Encoding: We first isolate the contribution of each sensing modality with language guidance disabled. DSM alone reaches 68.08% mIoU and RGB alone reaches 83.56%, whereas their combination reaches 84.41% — corresponding to the "w/o VLP Guidance" configuration in block A. Thus, under the same no-language setting, DSM contributes an additional 0.85 points on top of the optical input. The full model subsequently gains a further 1.96 points from vision–language guidance. This ordering is consistent with the feature separability shown in Fig. 12.

We additionally examine whether the simple DSM channel replication used by the full model limits performance.

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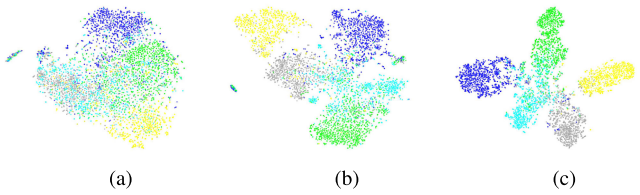


Fig. 12. t-SNE visualization of feature distributions on Vaihingen. Colors denote: impervious surfaces (gray), buildings (blue), low vegetation (cyan), trees (green), and cars (yellow). (a) DSM features exhibit limited class separability. (b) RGB features show improved but incomplete separation. (c) Decoded features form compact and well-separated clusters.

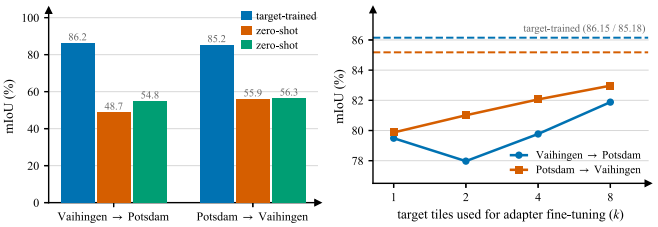


Fig. 13. Cross-city transfer between the ISPRS benchmarks. (a) Zero-shot IoU against a target-trained reference; the luminance control replicates gray in both domains to remove the IRRG/RGB mismatch. (b) Adapter fine-tuning on k target tiles, mean and s.d. over three draws, whose spread reaches 9.3 points at $k = 2$ and falls below 0.8 by $k = 8$.

Replacing replication with a learned $1 \rightarrow 3$ convolutional stem, initialized to reproduce the same input at the beginning of training, yields 86.16% mIoU compared with 86.37% for the default replication strategy. This result suggests that the simple input mapping is sufficient and that additional adaptation capacity is better allocated to the feature-level LMA modules.

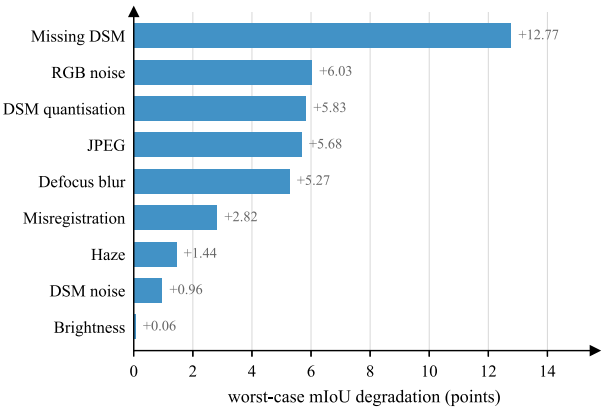


Fig. 14. Robustness of the Vaihingen model under test-time input degradation. Each bar is the worst case over the severities tested for that family, reported as the mIoU degradation *relative* to the clean input evaluated under the same inference configuration, so that all bars share a single reference point. Severities: RGB noise $\sigma \in \{0.02, 0.05, 0.1\}$, DSM noise $\sigma \in \{0.05, 0.1, 0.2\}$, DSM quantisation to $\{16, 8, 4\}$ levels, JPEG quality $\{50, 30, 10\}$, defocus $\{1, 2, 3\}$ pixels, haze $\{0.2, 0.4\}$, DSM–RGB misregistration $\{8, 16, 32, 64\}$ pixels, brightness scaling by $2^{\pm 0.5}$, and the DSM replaced by a constant.

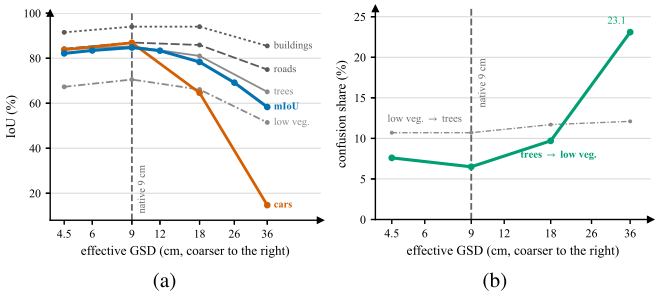


Fig. 15. Resolution transfer: the 9 cm Vaihingen model on resampled test tiles, as a change from the native result. (a) Per-class and mean IoU. (b) Low-vegetation/trees confusion.

F. Generalization and Robustness

1) Cross-City Transfer and Few-Shot Adaptation: Fig. 13 reports transfer between the two ISPRS benchmarks, which differ in geographic region, GSD (9 versus 5 cm), and optical channel convention (IRRG versus RGB). Zero-shot transfer is poor in both directions. A luminance control recovers at most 16% of the gap, indicating that the optical channel convention alone is not the dominant source of the transfer degradation. The remaining discrepancy likely reflects a combination of resolution, geometric, sensor, and geographic domain shifts.

Our primary claim therefore concerns efficient target-domain adaptation rather than zero-shot domain invariance. Fine-tuning on a single target tile recovers at least 82% of the gap to the target-trained reference, while eight target tiles recover 89% and 92% in the two transfer directions. These results support the interpretation of MAPLE's parameter efficiency primarily as an adaptation property rather than an inference-time efficiency property.

2) Robustness to Input Degradation: Fig. 14 evaluates the Vaihingen model under nine test-time degradation families, with all values reported relative to the clean input under the same inference configuration. Eight degradation families incur at most a 6.03-point mIoU loss at their most severe tested level. For example, DSM–RGB misregistration of 64 pixels, corresponding to approximately 5.8 m at the 9 cm GSD, reduces mIoU by 2.82 points, while strong additive DSM noise reduces it by 0.96 points.

The main exception is complete DSM removal, which causes a 12.77-point drop. For comparison, the training-time RGB-only ablation in Table V is only 0.85 points below the RGB + DSM no-language configuration (83.56% versus 84.41%). Because these results come from different trained models and evaluation protocols, they should not be interpreted as a direct numerical decomposition. Nevertheless, the large difference suggests that train–test modality mismatch is a major contributor to the severe degradation observed when DSM is removed only at inference time. Training with modality dropout or missing-modality augmentation may therefore improve robustness, which we leave for future work.

3) Resolution Transfer: The 9 cm model is evaluated without retraining on its own test tiles resampled between 4.5 and 36 cm (Fig. 15). The loss stays below 1.4 mIoU only within a narrowband around the training GSD—from 6 to 13 cm, i.e., roughly 1.5× finer to 1.4× coarser—and grows rapidly outside it, with the two directions failing differently. Coarsening removes small objects first: halving the resolution costs 6.4 points and quartering it 26.4, of which *cars* alone supply 69% and 55% while holding 0.84% of the pixels,

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TABLE VII

EFFICIENCY COMPARISON ON ISPRS VAIHINGEN AT A 256 × 256 INPUT. LATENCY IS MEASURED AT BATCH SIZE 1. FOR MAPLE, "TOTAL" INCLUDES THE FROZEN SAM3 VIT-L+ BACKBONE AND THE FROZEN REMOTECLIP VIT-B/32 ENCODER, BOTH OF WHICH PARTICIPATE IN EVERY FORWARD PASS

Method	Train. (M)↓	Total (M)↓	GFLOPs↓	Latency (ms)↓	Memory (MB)↓	mIoU (%)↑
PSPNet	46.72	46.72	49.03	**15.1**	3124	76.96
MAResU-Net	26.27	26.27	8.79	94.2	1908	79.89
UNetFormer	24.20	**24.20**	**6.04**	72.0	1980	81.97
RS^{3}Mamba	43.32	43.32	28.25	45.6	**1548**	82.78
TransUNet	93.23	93.23	32.27	92.5	3028	78.26
FuseNet	42.08	42.08	58.37	52.9	2284	78.71
V-FuseNet	44.17	44.17	60.36	59.1	2618	78.92
ESANet	34.03	34.03	7.73	96.0	1914	79.42
SA-Gate	110.85	110.85	41.28	100.0	3174	81.27
CMFNet	123.63	123.63	78.25	116.0	4058	81.44
MFTransNet	43.77	43.77	8.44	67.2	1549	81.61
CMGFNet	64.20	64.20	19.51	86.1	2463	82.26
FTransUNet	160.88	160.88	45.21	102.7	3463	84.23
FTransDeepLab	69.86	69.86	12.94	69.4	1624	83.87
MultiSenseSeg	60.46	60.46	66.28	40.6	2264	84.53
MFNet	111.28	748.3	594.39	158.3	6854	85.03
MAPLE (ours)	**19.51**	617.0	592.94	118.7	2601	**86.37**

whereas refining to 4.5 cm costs 2.6 points spread evenly across classes. The vegetation pair provides further insight into this behavior: trees → low vegetation confusion rises from 6.5% to 23.1% as the input is coarsened, whereas the reverse confusion changes much less. This asymmetry is consistent with the loss of fine-grained canopy texture under spatial coarsening and supports, without proving, our hypothesis that limited spatial resolution contributes to the grassland/forest confusion observed on MMHunan.

G. Model Efficiency Analysis

Table VII makes this accuracy–efficiency tradeoff explicit. Compared with MFNet, the strongest-performing baseline and the closest method in terms of foundation-model adaptation, MAPLE requires 5.7× fewer trainable parameters (19.51 versus 111.28 M), has fewer total parameters (617.0 versus 748.3 M), requires a comparable number of operations (592.94 versus 594.39 GFLOPs), and achieves lower single-image latency (118.7 versus 158.3 ms) while improving mIoU by 1.34 points. These results show that, relative to the closest foundation-model baseline, MAPLE substantially reduces task-adaptation cost while improving segmentation accuracy without increasing inference-time computation. The relatively high computational cost of both methods stems from the use of large pretrained encoders during inference, placing their FLOPs substantially above those of conventional hybrid architectures such as FTransUNet (45.21 GFLOPs).

The comparison with lightweight architectures reveals a different accuracy–efficiency regime. RS^{3}Mamba, for example, achieves 82.78% mIoU with 43.32 M total parameters, 28.25 GFLOPs, and 45.6 ms latency, whereas MAPLE achieves 86.37% mIoU at a higher inference cost, corresponding to a 3.59-point accuracy gain. Thus, MAPLE is not designed to minimize inference computation; instead, it targets a favorable balance between high segmentation accuracy and parameter-efficient adaptation of foundation models. Lightweight architectures retain an advantage under strict inference budgets, whereas MAPLE is more suitable when segmentation accuracy and adaptation efficiency are prioritized.

V. CONCLUSION

In this article, we presented MAPLE, a unified framework for multimodal remote sensing segmentation that adapts a frozen foundation backbone through lightweight modality-aware mechanisms and language-derived semantic priors. The proposed framework reconciles heterogeneous sensing modalities while preserving pretrained representations through parameter-efficient adaptation. Extensive experiments across three benchmarks demonstrate that MAPLE achieves state-of-the-art segmentation performance with only 19.51 M trainable parameters. The accompanying efficiency analysis further shows that MAPLE substantially reduces task-specific optimization cost relative to the closest foundation-model baseline, while its use of large frozen encoders entails a higher inference cost than lightweight task-specific architectures. These results position MAPLE primarily as an accuracy-oriented, parameter-efficient adaptation framework rather than a lightweight inference architecture. The added generalization analysis further shows that a single target tile recovers at least 82% of the gap to a target-trained model in cross-city adaptation, while eight of the nine tested degradation families cause at most a 6.03-point mIoU loss at their strongest evaluated severity. Complete DSM removal remains the principal failure case, causing a 12.77-point degradation. Comparison with the training-time modality ablations suggests that train–test modality mismatch is an important contributor to this failure, motivating missing-modality augmentation or modality dropout in future work.

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